

Fork-join parallelism

??/??/??

(drawing on materials by Dan Grossman)

Using multiple cores

- Essentially all computers now are multi-core (dual-core means 2 cores, quad-core means 4 cores, etc)
 - Each core can run all the operations needed to run a program
- Up to this point, our programs have always done a single thing at once so one can run on each core
- What if we wanted multiple cores to run a single program faster?
 - Need to split program into parts that can run on different cores

Sample problem: Summing an array

```
int simpleSum(int[] array, int low, int hi) {  
    int sum = 0;  
    for(int i=low; i < hi; i++)  
        sum = sum + array[i];  
    return sum;  
}
```

Splitting the sum into two parts

```
int sum(int[] array, int low, int hi) {  
    int left = simpleSum(array, low, (hi+low)/2);  
    int right = simpleSum(array, (hi+low)/2, hi);  
    return left + right;  
}
```

New conceptual keywords

(not real Java)

```
int sum(int[] array, int low, int hi) {
```

Allows child call (rest of line) to run in parallel with its parent

```
    int left = fork simpleSum(array, low, (hi+low)/2);
```

```
    int right = simpleSum(array, (hi+low)/2, hi);
```

```
    join;
```

Wait for **all** children to complete

```
    return left + right;
```

```
}
```

New conceptual keywords

(not real Java)

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int sum(int[] array, int low, int hi) {
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```
    int left = fork simpleSum(array, low, (hi+low)/2);
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```
    int right = simpleSum(array, (hi+low)/2, hi);
```

```
    join;
```

```
    return left + right;
```

```
}
```

Allows child call (rest of line) to run in parallel with its parent

Wait for **all** children to complete

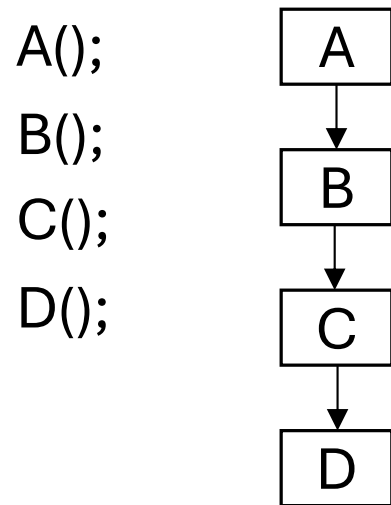
Hopefully, this

becomes these



time →

fork and join in pictures



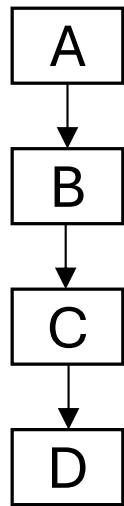
Each line must run in
the given order

Boxes are *strands* (chunk of
code we want to run without a
fork or join)

Arrows show *precedence
constraints*, relationship where
one strand must run before
another

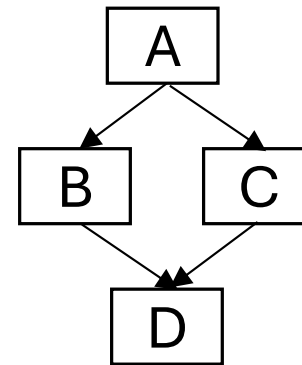
fork and join in pictures

A();
B();
C();
D();



Each line must run in
the given order

A();
fork B();
C();
join;
D();



Both B and C must run after A
and before D, but their
relationship is undefined

Which of the following could be printed by the code below?

```
fork System.out.print(1);  
System.out.print(2);  
fork System.out.print(3);  
join;  
System.out.print(4);
```

I. 3124

II. 2134

III. 1243

A. Only I.

B. Only II.

C. Only I. and II.

D. Only II. and III.

E. None of the above

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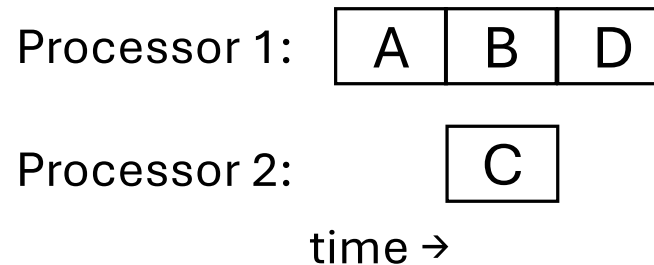
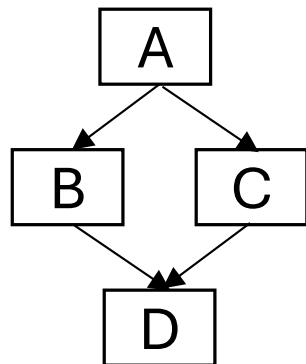
E. None of the above (I. and III.)

Running a precedence graph

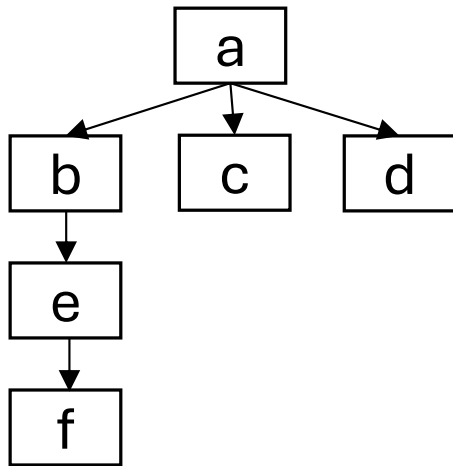
- Have a pool of cores/processors/threads (“processors”) that are available to run strands
- Each time unit, strands w/o unfinished predecessors are assigned to available processors and completed
- Typically assume strands represent equal amounts of work
- Processors are never voluntarily idle

Running a precedence graph

- Have a pool of cores/processors/threads (“processors”) that are available to run strands
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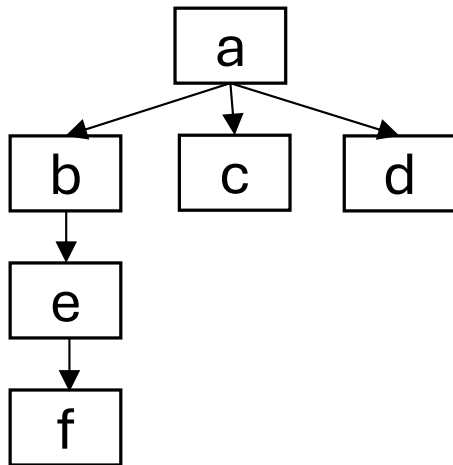


How long will this graph run on 2 processors?



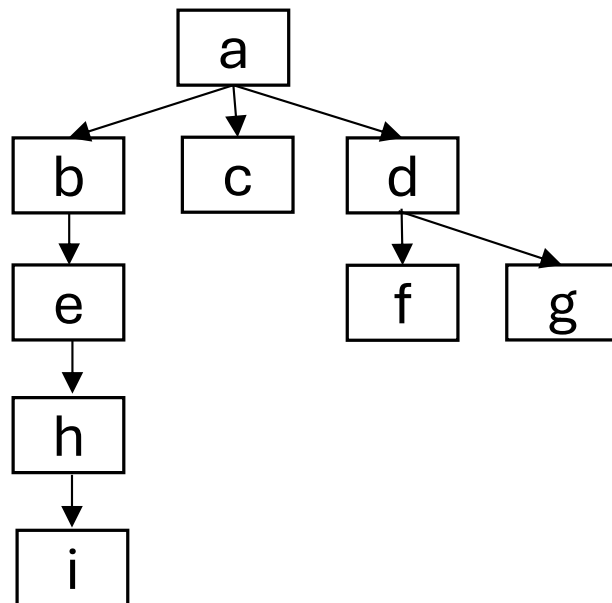
- A. Exactly 3 time units
- B. Exactly 4 time units
- C. Either 3 or 4 time units
- D. Either 4 or 5 time units
- E. None of the above

How long will this graph run on 2 processors?



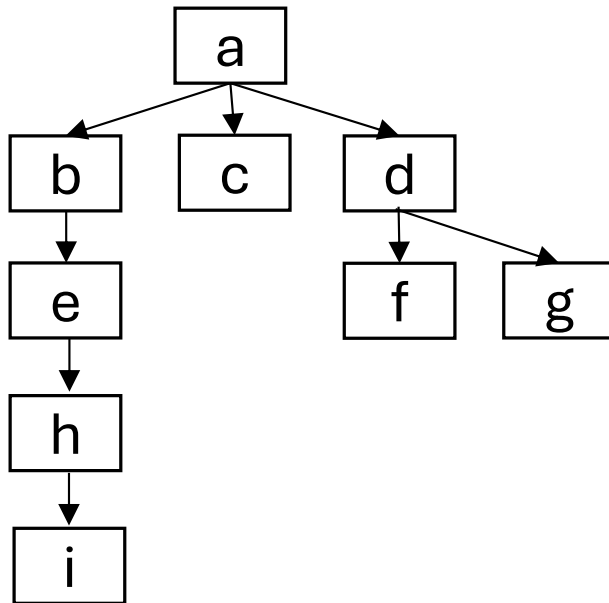
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How long will this graph run on 2 processors?



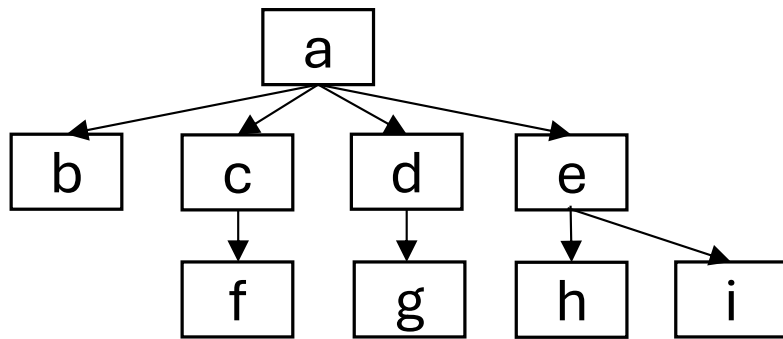
- A. Exactly 4 time units
- B. Exactly 5 time units
- C. Either 4 or 5 time units
- D. Either 5 or 6 time units
- E. None of the above

How long will this graph run on 2 processors?



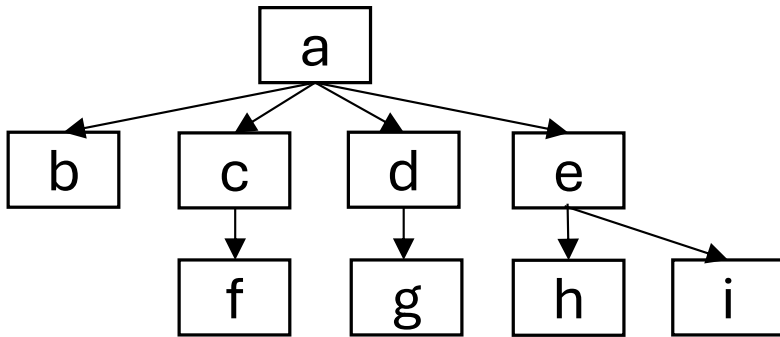
- A. Exactly 4 time units
- B. Exactly 5 time units
- C. Either 4 or 5 time units
- D. Either 5 or 6 time units
- E. None of the above
(between 5 and 7)

How long will this graph run on 2 processors?



- A. Exactly 4 time units
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- C. Either 4 or 5 time units
- D. Either 5 or 6 time units
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- D. Either 5 or 6 time units
- E. None of the above

Now we're trying to optimize 2 things

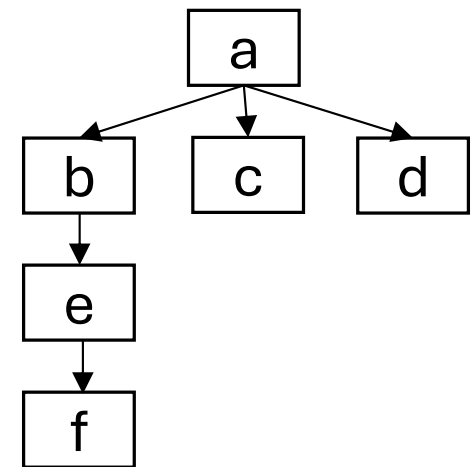
- Work = number of strands (denoted T_1)
- Span = length of longest path (denoted T_∞)

Now we're trying to optimize 2 things

- Work = number of strands (denoted T_1)
- Span = length of longest path (denoted T_∞)

What are the work and span of this graph?

- A. 3 and 4
- B. 4 and 4
- C. 6 and 3
- D. 7 and 3
- E. None of the above

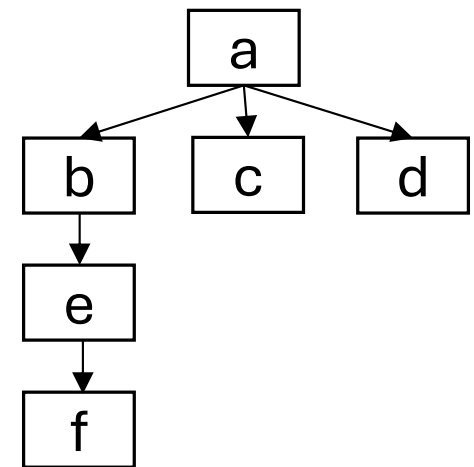


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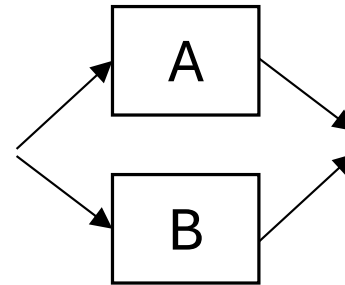
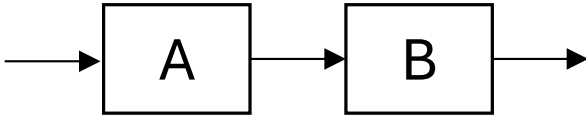
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- Span = length of longest path (denoted T_∞)

What are the work and span of this graph?

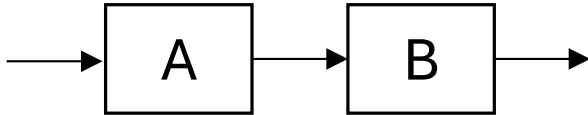
- A. 3 and 4
- B. 4 and 4
- C. 6 and 3
- D. 7 and 3
- E. None of the above (6 and 4)



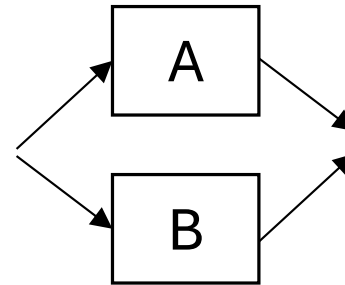
Combining subcomputations



Combining subcomputations

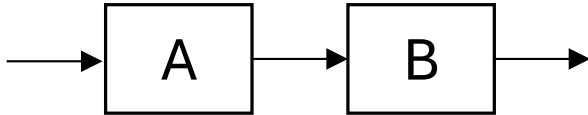


Work: $T_1(A) + T_1(B)$



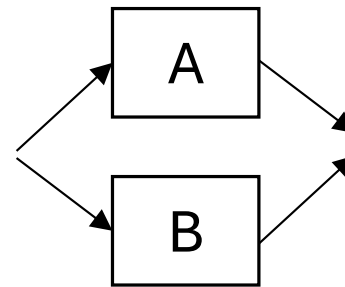
Work: $T_1(A) + T_1(B)$

Combining subcomputations



Work: $T_1(A) + T_1(B)$

Span: $T_\infty(A) + T_\infty(B)$



Work: $T_1(A) + T_1(B)$

Span: $\max\{ T_\infty(A), T_\infty(B) \}$

Using different (fixed) numbers of cores

```
int sum(int[] arr, int low, int hi) {  
    int p1 = (hi+2*low)/3;  
    int p2 = (2*hi+low)/3;  
    int s1 = fork simpleSum(array, low, p1);  
    int s2 = fork simpleSum(array, p1, p2);  
    int s3 = simpleSum(array, p2, hi);  
    join;  
    return s1 + s2 + s3;  
}
```

```
int sum(int[] arr, int low, int hi) {  
    int p1 = (hi+3*low)/4;  
    int p2 = (2*hi+2*low)/4;  
    int p3 = (3*hi+low)/4;  
    int s1 = fork simpleSum(array, low, p1);  
    int s2 = fork simpleSum(array, p1, p2);  
    int s3 = fork simpleSum(array, p2, p3);  
    int s4 = simpleSum(array, p3, hi);  
    join;  
    return s1 + s2 + s3 + s4;  
}
```

Generalizing to any number of cores

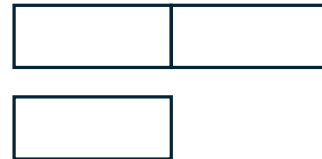
```
int sum(int[] arr, int low, int hi) {  
    int c = getNumCores();  
    int[] sums = new int[c];  
    for(int i=0; i < c; i++) {  
        int s = (i*hi + (c-i)*low) / c;  
        int e = ((i+1)*hi + (c-i-1)*low) / c;  
        sums[i] = fork simpleSum(array, s, e);  
    }  
    join;  
    return simpleSum(sums, 0, c);  
}
```

Generalizing to any number of cores

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        sums[i] = fork simpleSum(array, s, e);  
    }  
    join;  
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}
```

What if

... c is wrong? (the OS takes a core back)

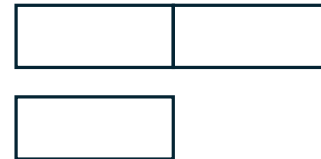


Generalizing to any number of cores

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    }  
    join;  
    return simpleSum(sums, 0, c);  
}
```

What if

... c is wrong? (the OS takes a core back)



... strands need different amounts of work?



Recursion to the rescue!

```
int sum(int[] array, int low, int hi) {  
    //base case  
    int left = fork sum(array, low, (hi+low)/2);  
    int right = sum(array, (hi+low)/2, hi);  
    join;  
    return left + right;  
}
```

Recursion to the rescue!

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int sum(int[] array, int low, int hi) {  
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    join;  
    return left + right;  
}
```

Creates many more strands than cores

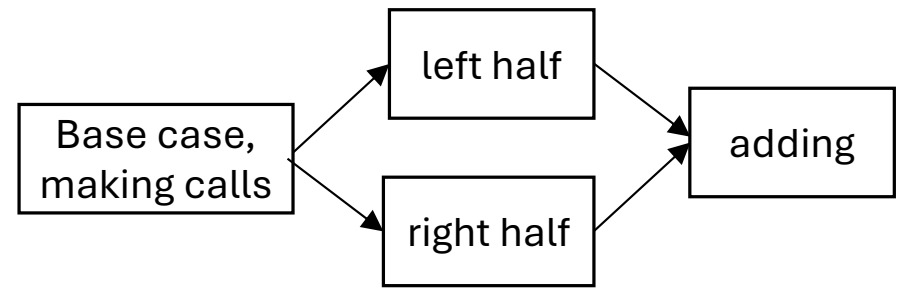
As each core finishes a strand, it gets another

Strands may have initial assignment, but any core can potentially take them (*work stealing*)

Precedence graph for this code

```
int sum(int[] array, int low, int hi) {  
    //base case
```

```
    int left = fork sum(array, low, (hi+low)/2);  
    int right = sum(array, (hi+low)/2, hi);  
    join;  
    return left + right;  
}
```



Precedence graph for this code

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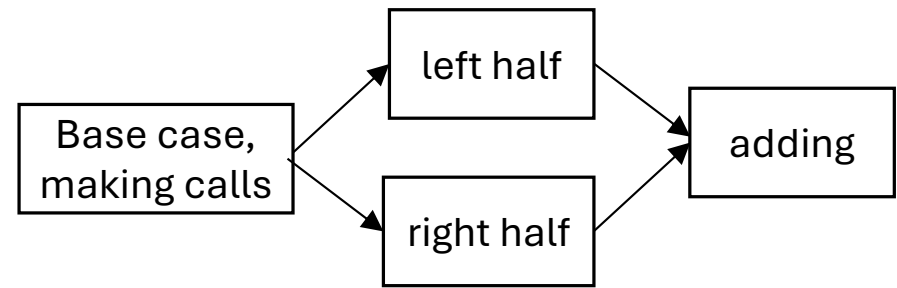
```
    int left = fork sum(array, low, (hi+low)/2);
```

```
    int right = sum(array, (hi+low)/2, hi);
```

```
    join;
```

```
    return left + right;
```

```
}
```



What is the span of this code? ($n=hi-low$)

A. $\Theta(1)$

B. $\Theta(\log n)$

C. $\Theta(n)$

D. $\Theta(n \log n)$

E. None of the above

Precedence graph for this code

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int sum(int[] array, int low, int hi) {  
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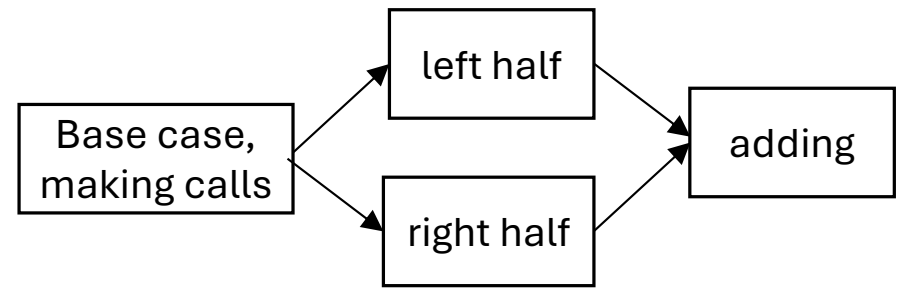
```
    int left = fork sum(array, low, (hi+low)/2);
```

```
    int right = sum(array, (hi+low)/2, hi);
```

```
    join;
```

```
    return left + right;
```

```
}
```



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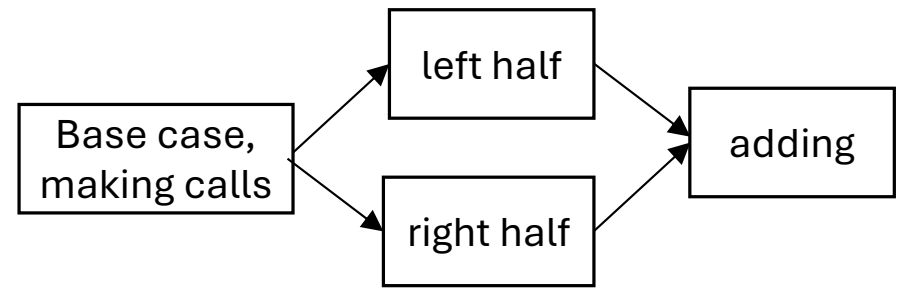
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    int left = fork sum(array, low, (hi+low)/2);
```

```
    int right = sum(array, (hi+low)/2, hi);
```

```
    join;
```

```
    return left + right;
```

```
}
```



What is the work of this code? ($n=hi-low$)

A. $\Theta(1)$

B. $\Theta(\log n)$

C. $\Theta(n)$

D. $\Theta(n \log n)$

E. None of the above

Applying this to a program

```
int sum(int[] array, int low, int hi) {  
    //base case
```

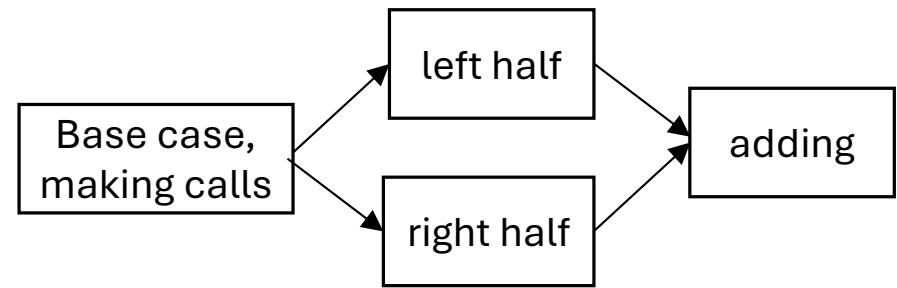
```
    int left = fork sum(array, low, (hi+low)/2);
```

```
    int right = sum(array, (hi+low)/2, hi);
```

```
    join;
```

```
    return left + right;
```

```
}
```



What is the work of this code? ($n=hi-low$)

A. $\Theta(1)$

B. $\Theta(\log n)$

C. $\Theta(n)$

D. $\Theta(n \log n)$

E. None of the above

What is the work and span of this code?

(n is hi-low)

```
int sum(int[] array, int low, int hi) {  
    int total = 0;  
    for(int i=low; i < hi; i++)  
        total += array[i];  
    return total;  
}
```

- A. $\Theta(\log n)$ and $\Theta(\log n)$
- B. $\Theta(\log n)$ and $\Theta(n)$
- C. $\Theta(n)$ and $\Theta(\log n)$
- D. $\Theta(n)$ and $\Theta(n)$
- E. None of the above

What is the work and span of this code?

(n is hi-low)

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int sum(int[] array, int low, int hi) {  
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```

- A. $\Theta(\log n)$ and $\Theta(\log n)$
- B. $\Theta(\log n)$ and $\Theta(n)$
- C. $\Theta(n)$ and $\Theta(\log n)$
- D. $\Theta(n)$ and $\Theta(n)$
- E. None of the above

Putting this into actual Java

```
int sum(int[] array, int low, int hi) {  
    ...  
}  
  
class Sum extends RecursiveTask<Integer> {  
    protected Integer compute() {  
        ...  
    }  
}
```

Use a class to represent a call to sum

Parameterized by the return type (must be a reference type)

Putting this into actual Java

```
int sum(int[] array, int low, int hi) {  
    ...  
}
```

Can't pass arguments into compute

Use constructor to get the information

```
class Sum extends RecursiveTask<Integer> {  
    int[] array;  int low;  int high;  
  
    public Sum(int[] array, int low, int hi) {  
        this.array = array; this.low = low; this.hi = hi;  
    }  
  
    protected Integer compute() {  
        ...  
    }  
}
```

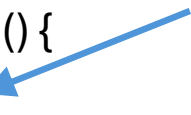
Putting this into actual Java

```
int sum(int[] array, int low, int hi) {  
    //base case  
  
    int left = fork sum(array, low, (hi+low)/2);  
    int right = sum(array, (hi+low)/2, hi);  
    join;  
    return left + right;  
}
```

...

```
protected Integer compute() {  
    if(hi - low < THRESHOLD)  
        return simpleSum(array, low, hi);  
    Sum left = new Sum(array, low, (hi+low)/2);  
    Sum right = new Sum(array, (hi+low)/2, hi);  
    left.fork();  
    return right.compute() + left.join();  
}
```

Only recurse to create
enough strands



fork for only one half
run right half recursively, but before waiting for left half



Putting this into actual Java: Starting off

```
sum = ForkJoinPool.commonPool().invoke(new Sum(array, 0, array.length));
```

Putting it all together

```
class Sum extends RecursiveTask<Integer> {  
    //attributes and constructor  
  
    protected Integer compute(){  
        if(hi - low < THRESHOLD) return simpleSum(array, low, hi);  
        Sum left = new Sum(array, low, (hi+low)/2);  
        Sum right = new Sum(array, (hi+low)/2, hi);  
        left.fork();  
        return right.compute() + left.join();  
    }  
}  
  
sum = ForkJoinPool.commonPool().invoke(new Sum(array, 0, array.length));
```

Putting it all together

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        left.fork();  
        return right.compute() + left.join();  
    }  
}
```

What if I wanted the minimum instead of the sum?

```
sum = ForkJoinPool.commonPool().invoke(new Sum(array, 0, array.length));
```

Putting it all together

```
class Sum extends RecursiveTask<Integer> {  
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        return right.compute() + left.join();  
    }  
}
```

What if I wanted the minimum
and its index?

```
sum = ForkJoinPool.commonPool().invoke(new Sum(array, 0, array.length));
```

Reductions

- A reduction takes a collection of values (e.g. an array) and combined them into one value
- Many useful reductions can be efficiently implemented using this framework
 - Sum, product, or combination using a boolean operator (and, or, xor)
 - Minimum or maximum (or minIndex/maxIndex)
 - Building a histogram (record of how many times each value occurs)
 - Counting number of values meeting a condition
 - IndexOf or lastIndexOf
 - Finding the longest sequence of matching values

How do we sum the values in an $n \times n$ array?

Bonus slides

- These are slides that I used after Lab 9, which asks the students to implement reductions including to find the k th smallest. Many in the class implemented the method using library sorting routines so I used these slides to show them what I intended.
- I plan to adjust the Lab 9 handout so these hopefully won't be necessary, but they're included in case they're useful. (Perhaps if you want to talk about this reduction without doing the lab.)

Finding minimum

```
int min = nums[0];  
for(int i=1; i < nums.length; i++)  
    if(nums[i] < min)  
        min = nums[i];
```

Finding second smallest

```
int first = nums[0];           //smallest so far
int second = nums[1];         //second smallest so far
if(first > second)
    swap first and second;
for(int i=2; i < nums.length; i++) {
    if(nums[i] < second) {
        second = nums[i];
        if(second < first)
            swap first and second;
    }
}
```

Finding k^{th} smallest

```
int[] smallest = new int[k];           //k smallest values so far
...
for(int i=k; i < nums.length; i++) {
    if(nums[i] < smallest[k-1]) {
        int j = k-1;
        while((j > 0) && (nums[i] < smallest[j-1])) {
            smallest[j] = smallest[j-1];
            j--;
        }
        smallest[j] = nums[i];
    }
}
```

Finding k^{th} smallest

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int[] smallest = new int[k];           //k smallest values so far
...
for(int i=k; i < nums.length; i++) {
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        int j = k-1;
        while((j > 0) && (nums[i] < smallest[j-1])) {
            smallest[j] = smallest[j-1];
            j--;
        }
        smallest[j] = nums[i];
    }
}
```

What is the running time of this?

- A. $\Theta(n)$
- B. $\Theta(n+k)$
- C. $\Theta(n \log k)$
- D. $\Theta(n^2)$
- E. None of the above

Finding k^{th} smallest

```
int[] smallest = new int[k];           //k smallest values so far
```

```
...
```

```
for(int i=k; i < nums.length; i++) {
```

```
    if(nums[i] < smallest[k-1]) {
```

```
        int j = k-1;
```

```
        while((j > 0) && (nums[i] < smallest[j-1])) {
```

```
            smallest[j] = smallest[j-1];
```

```
            j--;
```

```
        }
```

```
        smallest[j] = nums[i];
```

```
    }
```

```
}
```

What is the running time of this?

A. $\Theta(n)$

B. $\Theta(n+k)$

C. $\Theta(n \log k)$

D. $\Theta(n^2)$

E. None of the above $\Theta(nk)$